Equilibrium vacancy concentration

A distinctive characteristic of point defects is that they can be present at substantial concentrations under conditions of thermodynamic equilibrium. They are sometimes called "thermodynamically reversible defects" - they can be controlled by thermodynamic parameters - T and P.

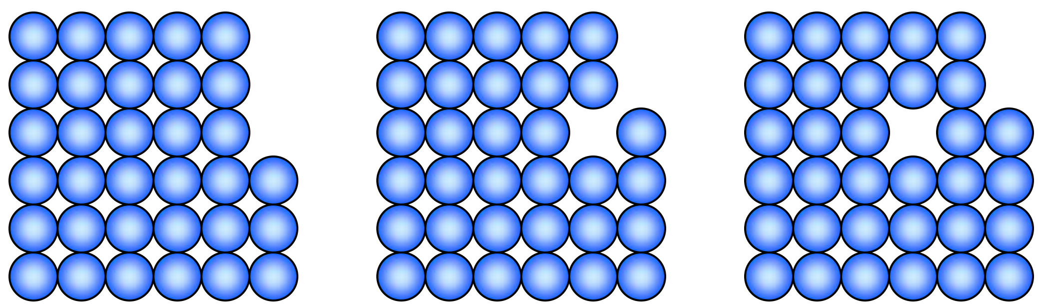
Thermodynamic equilibrium under conditions of constant P and T: $\boldsymbol{G = H - TS} \rightarrow min$

To find the equilibrium vacancy concentration, we have to find the change in G as a result of the introduction of n vacancies into a crystal with N lattice sites:

$$\Delta G = G^n - G^0 = \Delta H_f^n - T\Delta S^n$$

$$\Delta H_f^n = n\Delta h_f$$

Δh_f is the enthalpy of vacancy formation - lattice relaxation makes it substantially (~3-4 times) smaller than the heat of vaporization



vacancy formation by diffusion from the surface

E.g., for Au: $\Delta h_f \approx 0.97$ eV, latent heat of vaporization $\Delta h_s \approx 2.6$ eV, cohesive energy $E_c \approx 3.8$ eV

What is entropy?

"Any method involving the notion of entropy, the very existence of which depends on the second law of thermodynamics, will doubtless seem to many far-fetched, and may repel beginners as obscure and difficult of comprehension."

Willard Gibbs

Introduced in the 2nd law of thermodynamics to quantify irreversibility of a process:

2nd Law: There exist a state function, the entropy S, which for all reversible processes is defined by $dS = \delta q_{rev}/T$ and for all irreversible processes is such that $dS > \delta q/T$, or in general, $dS \geq \delta q/T$.

The entropy of a system plus its surroundings (together forming "the universe" – an isolated system - $\delta q = 0$) never decreases and increases in any irreversible process.

Physical interpretation of entropy: In statistical thermodynamics entropy is defined as a **measure of randomness or disorder**.

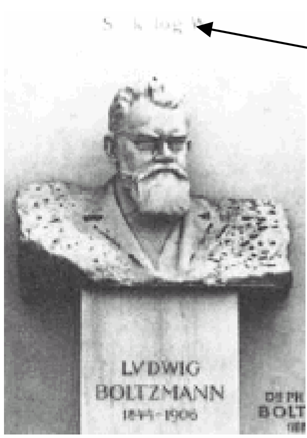
How to quantify "disorder"?

What is entropy?

How to quantify "disorder"? - microstates and macrostates

A **macroscopic state** of a system can be described in terms of a few macroscopic parameters, e.g. P, T, V. The system can be also described in terms of **microstates**, e.g. for a system of N particles we can specify coordinates and velocities of all atoms.

The 2nd law can be stated as follows: **The equilibrium state of an isolated system is the one in which the number of possible microscopic states is the largest.**



$$S = k_B \ln \Omega$$

Ω is the number of microstates, k_B is the Boltzmann constant (it was first introduced in this equation), and S is the entropy.

The entropy is related to the number of ways the microstate can rearrange itself without affecting the macrostate.

The 2nd law can be restated again: An isolated system tends toward an equilibrium macrostate with maximum entropy, **because** then the number of microstates is the largest.

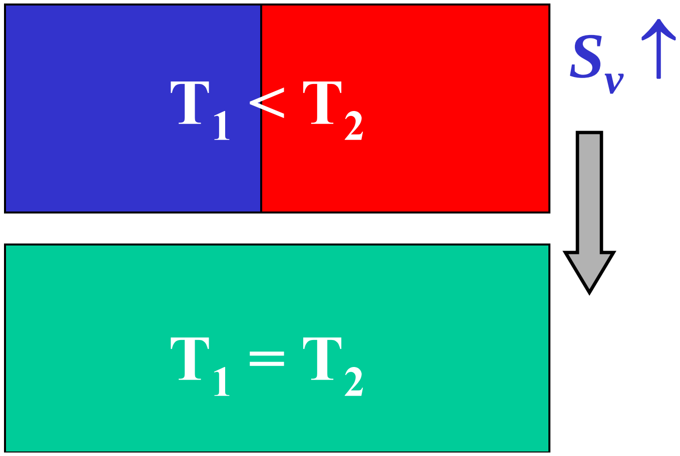
Configurational entropy and thermal (vibrational) entropy

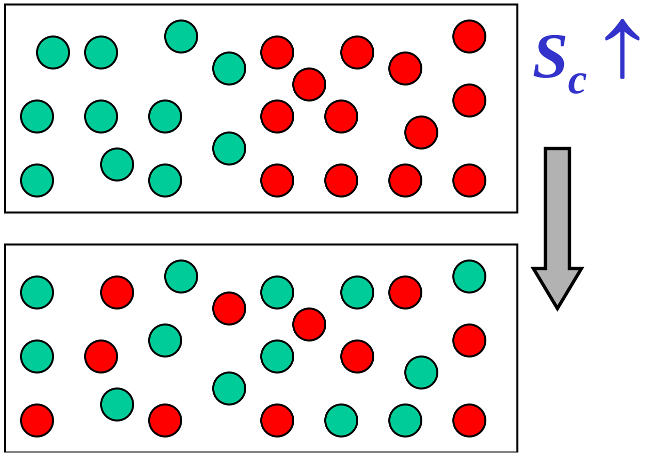
Vibrational entropy S_v: Entropy associated with lattice vibrations (the number of microstates Ω can be thought as the number of ways in which the thermal energy can be divided between the atoms).

The vibrational entropy of a material increases as the temperature increases and decreases as the cohesive energy increases.

The vibrational entropy plays important role in many polymorphic transitions. With increasing T, the polymorphic transition is from a phase with lower S_v to the one with lower vibrational frequencies and higher S_v, e.g. fcc → bcc ($\Delta H_{tr} > 0$ can be compensated by $-T\Delta S < 0$).

Configurational entropy S_c: Entropy can be also considered in terms of the number of ways in which atoms themselves can be distributed in space, e.g. entropy of mixing in quasi-chemical model.





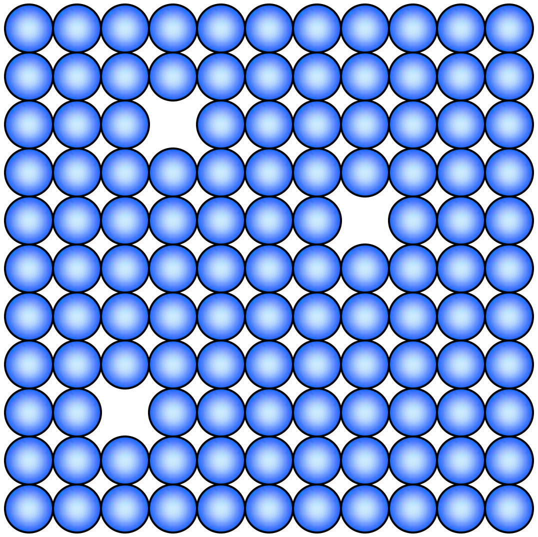
Equilibrium vacancy concentration

$$\Delta G = G^n - G^0 = \Delta H_f^n - T\Delta S^n$$

$$\Delta H_f^n = n\Delta h_f \qquad \Delta S^n = \Delta S_c^n + \Delta S_v^n \qquad \Delta S_v^n = n\Delta s_v$$

Δs_v is change in the vibrational entropy due to the introduction of one vacancy - related to changes in the vibrational frequencies of atoms surrounding the vacancy

$$\Delta G = G^n - G^0 = n(\Delta h_f - T\Delta s_v) - T\Delta S_c^n$$



$$\Delta S_c^n = S_c^n - S_c^0 \qquad S_c^n = k_B \ln(\Omega_n)$$

$$n = 0: \; S_c^0 = k_B \ln(1) = 0$$

$$n = 1: \; S_c^1 = k_B \ln(N)$$

$$n = 2: \; S_c^2 = k_B \ln[1/2\, N(N-1)] >> S_c^1$$

$$n = 3: \; S_c^2 = k_B \ln[1/6\, N(N-1)(N-2)] >> S_c^2$$

… in general, the number of distinct configurations for *n* vacancies is

$$\frac{1}{\mathrm{n!}}\mathrm{N}(\mathrm{N}-1)(\mathrm{N}-2)\ldots(\mathrm{N}-\mathrm{n}+1) = \frac{\mathrm{N!}}{\mathrm{n!}(\mathrm{N}-\mathrm{n})!}$$

Equilibrium vacancy concentration

$$S_c^n = k_B \ln \frac{N!}{n!(N-n)!} \qquad \Delta G = n(\Delta h_f - T\Delta s_v) - T\Delta S_c^n$$

Equilibrium concentration of vacancies can be found from: $\left.\frac{\partial \Delta G}{\partial n}\right|_{n=n_{eq}} = 0$

Let's first consider $\frac{\partial S_c^n}{\partial n}$ using Stirling formula for big numbers: $\ln N! \approx N \ln N - N$

$$\frac{\partial S_c^n}{\partial n} = k_B \frac{\partial}{\partial n}\{\ln N! - \ln n! - \ln(N-n)!\} \approx k_B \frac{\partial}{\partial n}\{N \ln N - N - n\ln n + n - (N-n)\ln(N-n) + (N-n)\} =$$

$$k_B \frac{\partial}{\partial n}\{N \ln N - n \ln n - (N-n)\ln(N-n)\} = k_B\left\{-\ln n - 1 + \ln(N-n) - \frac{(N-n)}{(N-n)}(-1)\right\} = k_B \ln\left(\frac{N-n}{n}\right) =$$

$$k_B \ln\left(\frac{N}{n} - 1\right) \approx -k_B \ln\left(\frac{n}{N}\right)$$

$$\left.\frac{\partial \Delta G}{\partial n}\right|_{n=n_{eq}} = \left.\frac{\partial}{\partial n}\left\{n(\Delta h_f - T\Delta s_v) - TS_c^n\right\}\right|_{n=n_{eq}} = \Delta h_f - T\Delta s_v + k_B T \ln(n_{eq}/N) = 0$$

$$\frac{n_{eq}}{N} = \exp\left(\frac{\Delta s_v}{k_B}\right)\exp\left(-\frac{\Delta h_f}{k_B T}\right)$$

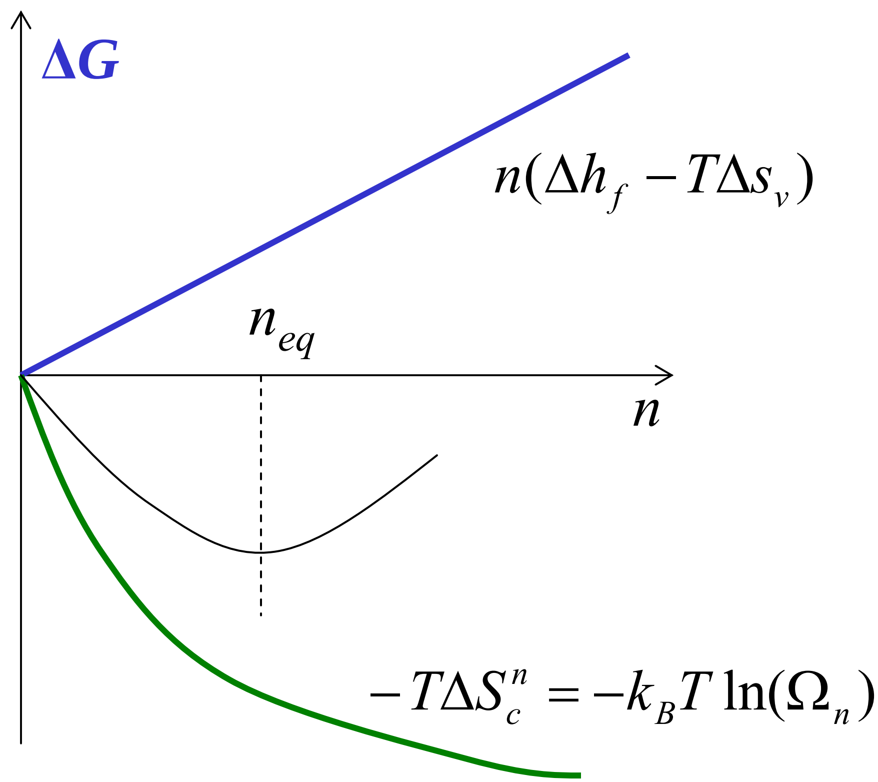
Equilibrium vacancy concentration

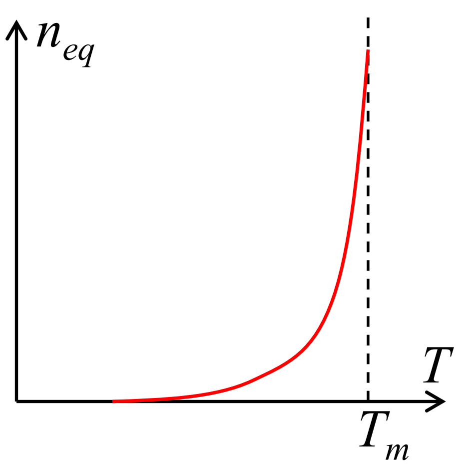
$$\frac{n_{eq}}{N} = \exp\left(\frac{\Delta s_v}{k_B}\right)\exp\left(-\frac{\Delta h_f}{k_B T}\right)$$

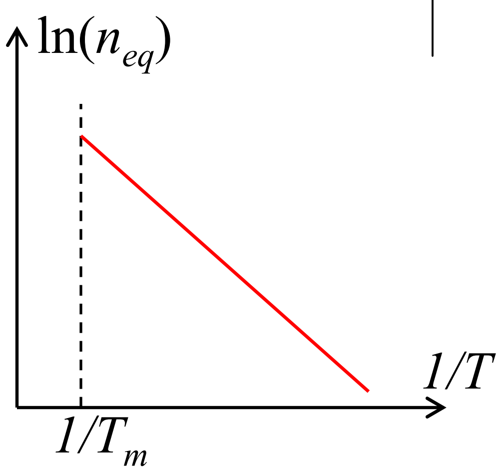
The equilibrium vacancy concentration is defined by the balance between $n(\Delta h_f - T\Delta s_v)$ and $-T\Delta S_c^n$

$$\left.\frac{\partial(-T\Delta S_c^n)}{\partial n}\right|_{\frac{n}{N}\to 0} \approx k_B T \ln\left.\left(\frac{n}{N}\right)\right|_{\frac{n}{N}\to 0} \to -\infty$$

steep dependence of the entropy term at small *n/N*
→ *defect-free crystals do not exist* (nanoparticles?)



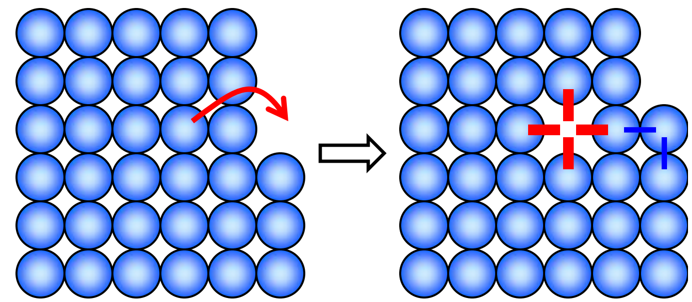




Energy of vacancy formation Δh_f

Rough estimation of Δh_f based on an assumption that cohesive energy E_c is equal to the sum of bond energies between adjacent atoms.

By moving an atom to the surface: (1) bonds of an atom that was at the vacancy site are broken (adding E_c), (2) some of the bonds of the atoms surrounding the vacancy are broken (adding another E_c), (3) atom is moved to the surface and restores some of the bonds (removing E_c), (4) atoms around the vacancy are relaxed (reducing energy by ΔE_R):



$$\Delta h_f = 2E_c - E_c - R = E_c - \Delta E_R$$

The accuracy of experimental and theoretical evaluation of Δh_f is limited, with substantial variability in reported data.

$$E_c = \frac{1}{N_{atoms}} \left(E_{solid} - \sum_i E_i^{atom} \right)$$

material	Al	Au	Cu	Ni	α-Fe	W	Si	Ar
Δh_f (eV)[a]	0.68	0.97	1.22	1.6	1.5	3.7	2.1[c] - 3.6[d]	0.06-0.09[e]
E_c (eV)[b]	3.39	3.81	3.49	4.44	4.28	8.9	4.63	0.08
T_m (K)[a]	933	1336	1356	1726	1808	3653	1687	84

[a] Allen & Thomas
[b] Kittel, *Introduction to Solid State Physics*
[c] Bracht et al., PRL 91, 245502, 2003
[d] Dannefaer, Mascher, Kerr, PRL **56**, 2195, 1986
[e] Schwalbe, PRB **14**, 1722, 1976

Δh_f is mostly the energy of electron density rearrangement around the vacancy, with the elastic energy making relatively small contribution